

M. Bala Vighnesh Reddy

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SUMMARY

AI/ML engineer with experience in **computer vision**, **LLM systems**, **RAG**, and **efficient model training/deployment**. Interested in **RL for LLMs**, robotics, vision models, and compact models. Built production-oriented ML pipelines across segmentation, pose estimation, medical imaging, Text-to-SQL reasoning, document querying, Vision AI APIs, and **agentic applications using MCP and Groq**.

EDUCATION

Indian Institute of Information Technology, Manipur

Nov 2021 -- Apr 2025

Bachelor of Technology in Computer Science and Engineering

WORK EXPERIENCE

Computer Vision Intern

Aug 2025 -- Feb 2026

BigThinX, Bengaluru

- Built a fashion-domain computer vision pipeline combining **U2-Net**, **P3M-Net**, **DW Pose**, and **MediaPipe** for foreground segmentation and pose estimation on 2-view image inputs.
- Tested the **SMPL parametric body model** for 3D body mesh estimation from sparse 2D observations, and used the findings to refine the pipeline direction for loose-fitting garments.
- Enhanced low-resolution and low-light inputs using **LLFormer** and **KANT**, improving the existing pipeline performance by **30%**; deployed the system on a VM with a **Gradio** interface for real-time inference.

AI/ML Intern

Jan 2025 -- May 2025

IF MedTech, Navi Mumbai

- Fine-tuned a **Vision Transformer (ViT)** on limited medical imaging data and quantized it to **INT8**, achieving **89% accuracy** with reduced model size and inference time.
- Built medical image workflows using **Faster R-CNN** for anatomical region detection and **DeepLabV3** for semantic segmentation, reaching **0.64 IoU** on limited annotated data.
- Implemented a medical **RAG application** over a reference book using vector embeddings and **Llama3-8B** for context-grounded query answering.

PROJECTS

Text-to-SQL Research : Qwen3-4B-Instruct, DeepSeek, QLoRA, GRPO, Tinker API

- Distilled reasoning from **DeepSeek** into **Qwen3-4B** using **LoRA**, generating **9,000+** Chain-of-Thought SQL examples for Spider and BIRD.
- Implemented **execution-based reward functions** for **GRPO**, using SQL correctness as a verifiable reinforcement learning signal and improving performance by **10%** over baseline.
- Achieved **86.7% pass@4** on Spider dev and **70.7% pass@4** on BIRD dev using compute-efficient **4-bit QLoRA** training.

ml-model-registry : TypeScript, React, FastAPI, MongoDB, Docker, YOLOv8n

[GitHub](#)

- Built a full-stack **Vision AI API platform** exposing image classification, object detection, OCR, and segmentation through developer-friendly REST APIs.
- Integrated a Python **FastAPI ML inference worker** with lightweight models including **YOLOv8n**, **ResNet50**, **EasyOCR**, and **YOLOv8n-seg**.
- Added a React dashboard for model catalog browsing, live inference testing, API key management, usage tracking, and billing workflows; containerized services with **Docker Compose**.

AutoMCQGen : Llama2-7B, GPT-3.5, NLP, LMQL

[GitHub](#)

- Built a multiple-choice question generator using **Llama2-7B** and **LMQL**, constraining model outputs to strict schemas and subject-aligned formats.
- Designed an NLP preprocessing pipeline with rule-based splitting and topic modeling to convert long academic documents into logical, topic-aware chunks.
- Used **GPT-3.5** as an LLM-as-judge evaluator for correctness and relevance, achieving **87% content alignment**.

Document Querying RAG System : LlamaIndex, Node.js, React, MongoDB

[GitHub](#)

- Developed a **Retrieval-Augmented Generation** system using **LlamaIndex** to query large document collections with context-grounded responses.
- Created **Node.js REST APIs** for document upload, indexing, and querying, backed by **MongoDB** and a React frontend with expandable chat threads.
- Implemented secure authentication and session management to protect user data.

TECH SKILLS

- Languages & Frameworks**: Python, C/C++, TypeScript, JavaScript, FastAPI, Flask, Node.js, React.js
- AI/ML**: PyTorch, HuggingFace, Transformers, LLMs, LoRA, QLoRA, RAG, VectorDB, LangChain, LlamaIndex, UnSloth
- Computer Vision**: ViT, Faster R-CNN, DeepLabV3, YOLOv8, EasyOCR, MediaPipe, U2-Net, P3M-Net, segmentation, detection, pose estimation
- MLOps & Tools**: Docker, Docker Compose, Git, GitHub, Gradio, Ollama, OpenRouter, MongoDB, MySQL, GCP

CERTIFICATIONS

- **Generative AI with Large Language Models** -- DeepLearning.AI (2024)
- **Neural Networks and Deep Learning** -- DeepLearning.AI / Coursera (2024)